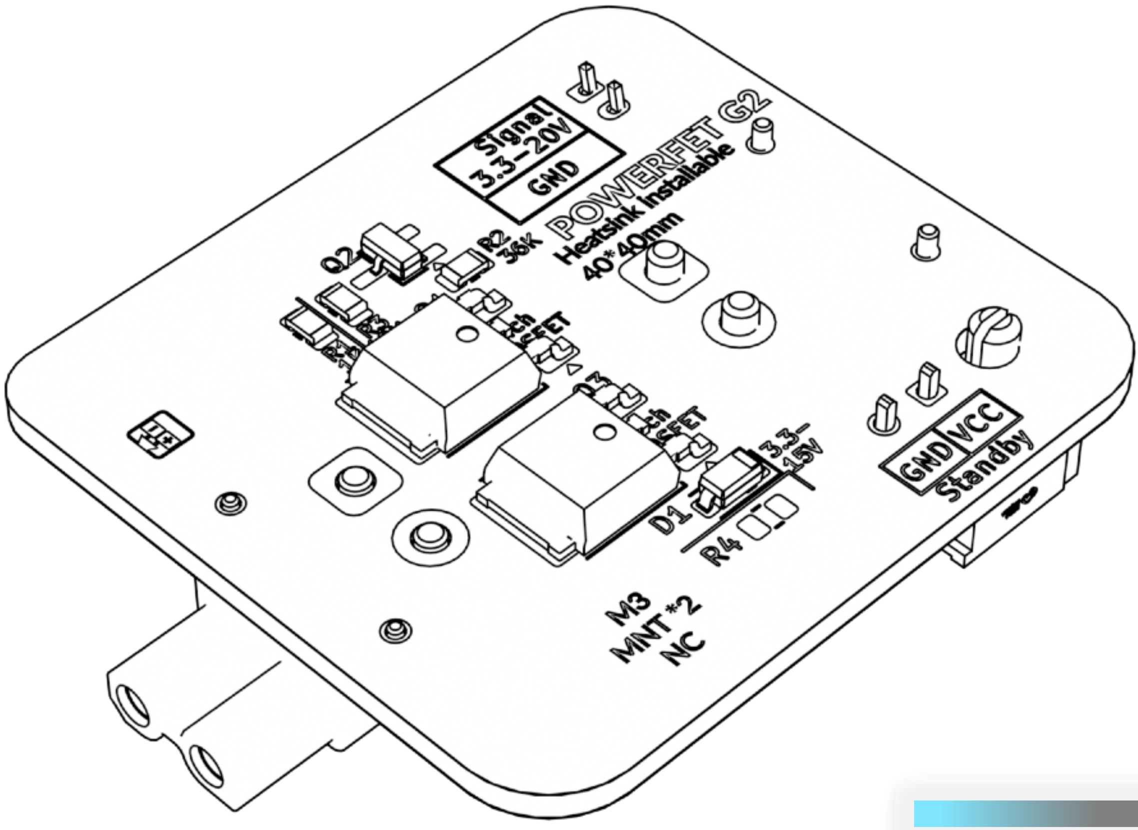
OIWP POWERFET
Generation 2
Assembly guide
REV 2026-08-20



OITSWILLIAM PANG Hardware

Contents

Before getting started

You are about to read the manual of OIWP POWERFET.

Warning

⚠ Danger: unless if it only runs on DC and only has a DC input, otherwise risk of electric shock

- Even though you can build with this mod from start, if you want to wire, or re-wire, you need to make sure and check that the mains is unplugged.
- Failure doing so which is unplugging the mains power may cause electrocution or even death.
- When in doubt, please don't touch the wires connected to the power supply or SSR.

Other notes

- **POWERFET is not a true solid-state relay / TRIAC replacement, it is just an unit with two-side MOSFETs with maximum gate-source voltage incompatible with AC loads. Connection of POWERFET to AC load can cause permanent damage to electronics.**
- **If the target unit's use of switches (such as SSR) that fail-close, POWERFET may not be used.**

About OIWP POWERFET

OIWP POWERFET is a MOSFET board that can switch power of printer controllers, because modern 3D printers should behave like modern computers. It fixes the weaknesses of common MOSFET switch boards' only negative-side and patches the traditional relay's power draw. POWERFET G2 can draw peak currents of up to 70A on 30V variant and 50A on 60V variant. Since the circuit is (rather) simple, solder bump on copper is added as a workaround of denser copper, making the unit budget-friendly. The idea of POWERFET was formed in late 2023 but wasn't seeing lights of days to the public until too late.

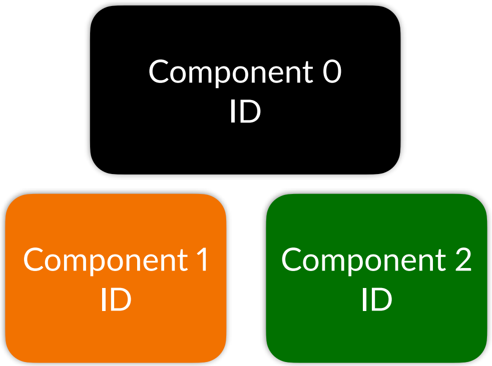
Contribution

This project can be contributed through GitHub. You can visit, contribute and report new issues here.

Technical support

You can contact Oitswilliam here.

Guide and legend



- highlighted materials (mostly showing components.)

Colors will vary to prevent misinterpretation.

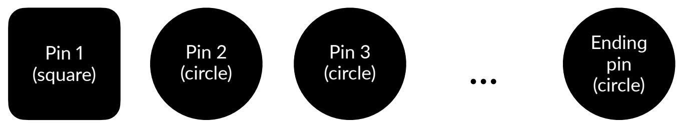
Component guide

KΩ - kiloohm (1KΩ = 1000Ω)
R = resistor
C = capacitor
D = diode
J = connector
JP = jumper

Tips may use additional colors.

Pin legend

The first pin are always in the square pad (with rounded corners), while the second pin and the rest are in the circle pads. However these cannot be easily observed if they are covered, especially if the front face was covered. Sometimes there are people who say the ground are square but that isn't right, unless if they are the first pins.



Additional maker tools

To get started building the project, you may use the following tools:

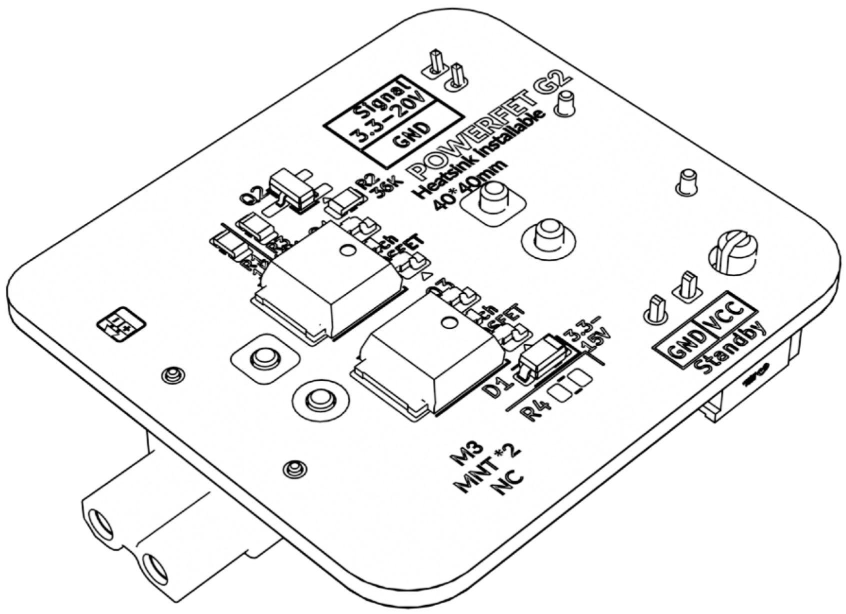
- Soldering iron
- Hot plate, if available, otherwise a pair of tweezers is needed.

Part printing

The cover for POWERFET is recommended to be printed in materials that are fire-redundant. **Bold text** indicates the recommended value and *italic text* indicates on what setting / value it needs to set.

FDM / FFF or SLS	Print material: PA, fire-redundant ABS or fire-redundant PC. Minimum being ABS or ASA.
Layer height of **0.2mm,** initial layer height can be **0.2mm or higher.**	*Nozzle and extrusion width* must be **forced 0.4mm; if using 0.4mm nozzle**
1.6mm *wall width*, or **4** *wall counts.*	*Infill* must be **at least 15%.**
At least 4 *solid bottom and top layers.*	The best *print speed* is up to **120mm/s**. The speed might be reduced by the hot end flow rate.
Supports are **recommended for main skirt parts.** Use **tree or organic supports** if available.	*Other settings might be depended on your printer.* *Follow it's warnings for proper handling.*

Assembly


Signal
3.3-20V
GND
POWERFET G2
Heatsink Installable
40*40mm
Q2
R2
36K
3.3-
15V
D1
R4
GND
VCC
Standby
M3
MNT*2
NC

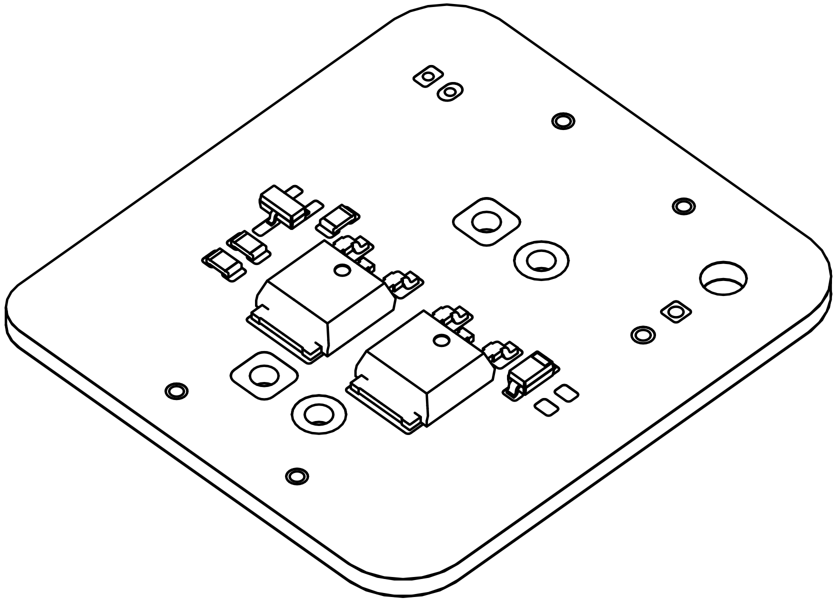
Board requirements

This shows the recommend set of settings, spec and parameters that is needed when sourcing and assembling POWERFET G2. PCBA for POWERFET G2 will become available within foreseeable future.

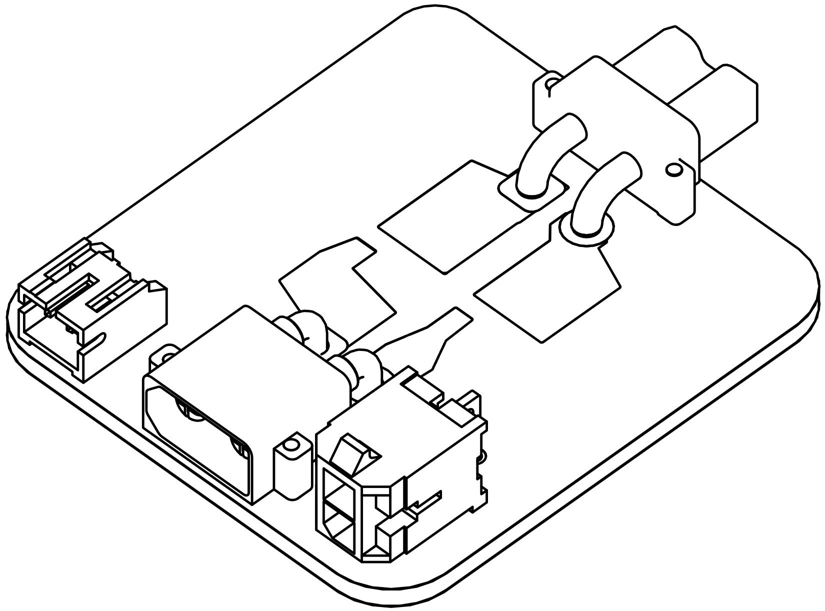
Name \ parameter	Minimum	Recommended
PCB color	Any color	
PCB thickness	1.6mm	
PCB conductor	HASL leaded	HASL lead-free
Device precision	Resistor: ±1%, capacitor: ±10%	
Solder wire / paste	Either leaded or lead-free Wire: up to 0.8mm Solder must withstand in at least 180˚C	Lead-free Wire: up to 0.8mm Solder must withstand in at least 180˚C
Stencil	Optional	

PCB assembly

This unit works best by placing surface-mount components and then through-hole components.
You may get ready by adding the solder paste to the board with the help of the stencil.
If you are using the hot plate, you will follow the the placement guide before it can be placed in the hot plate, this only applies to surface-mount components.

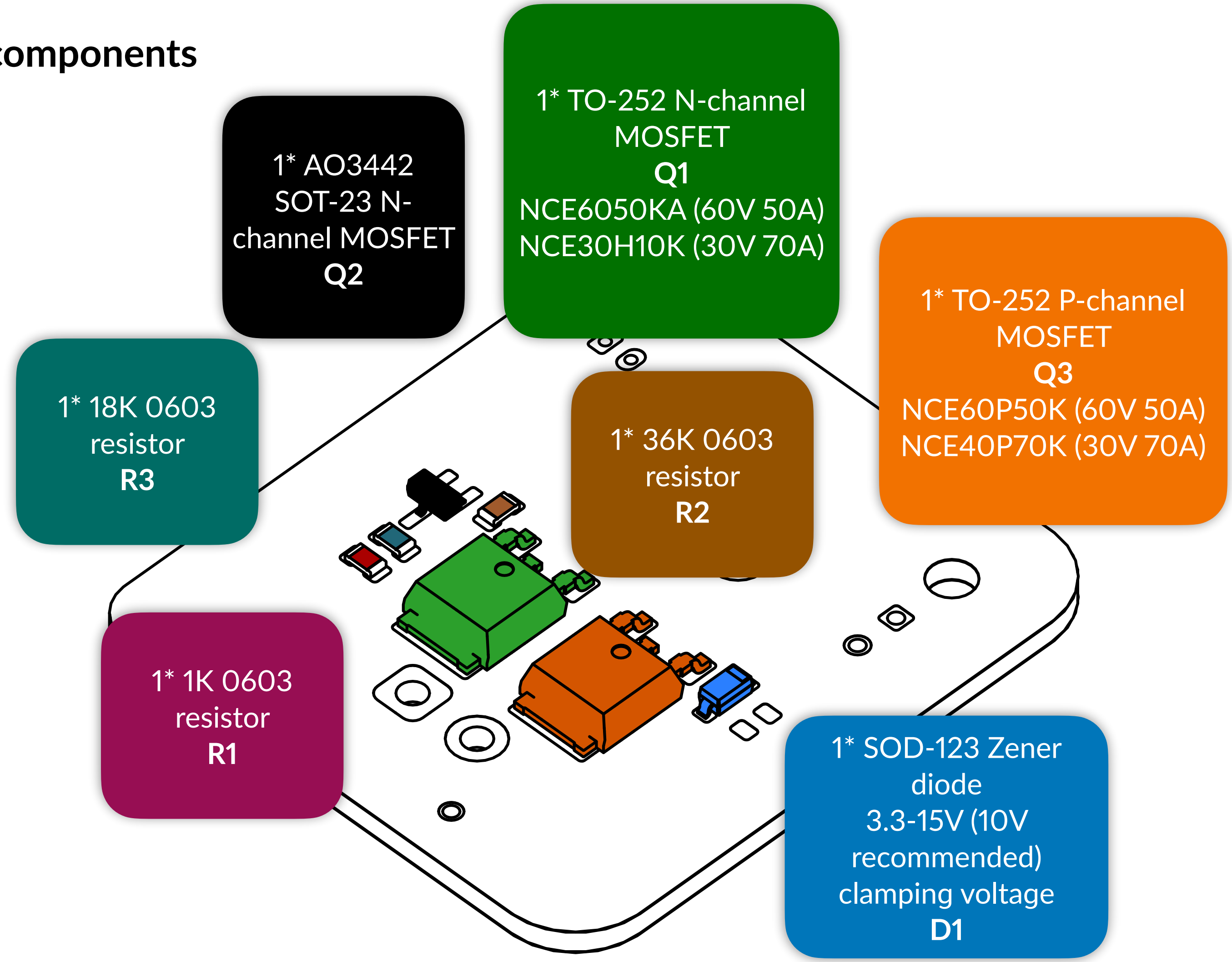


Chapter 1. Surface-mount components

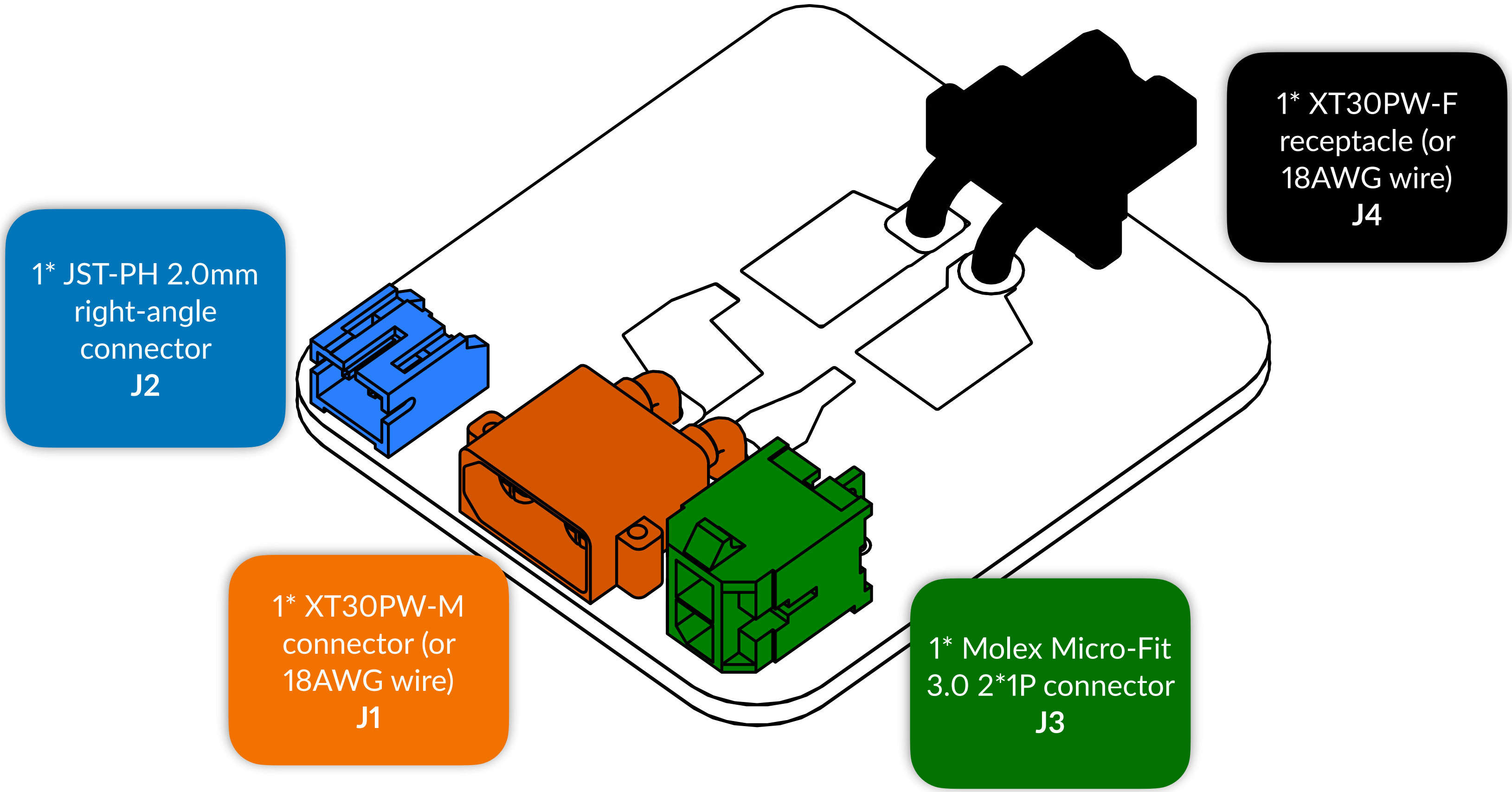


Chapter 2. Through-hole components

Surface-mount components

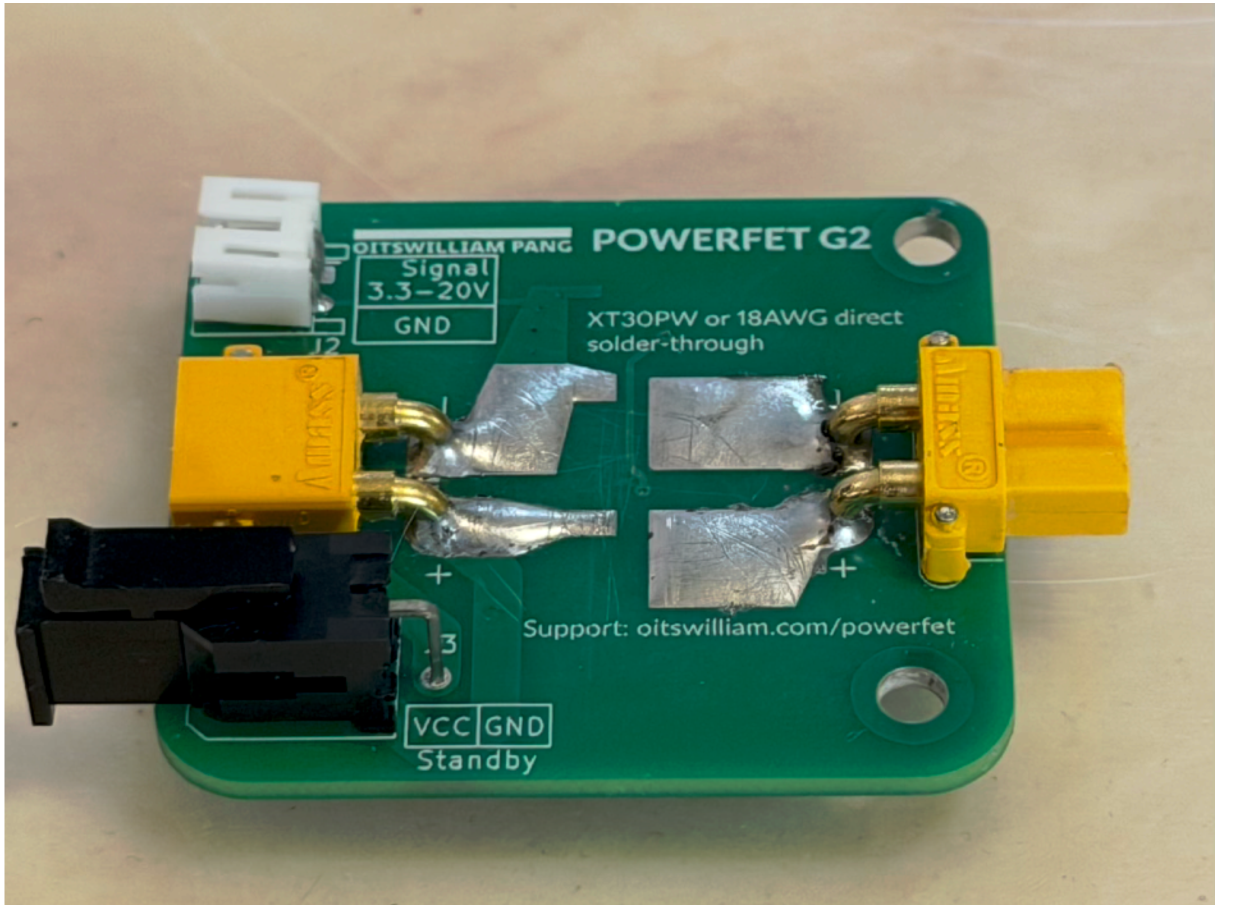


Through-hole components



Users can opt to solder the wires directly to the pads instead of XT30. Check the polarity when soldering directly.
After attaching the components and soldering these, a solder bump on the exposed copper layer is needed. Apply solder to the exposed copper that goes from XT30 receptacle to the MOSFET and the plug. Do not bridge the copper while adding the bump.

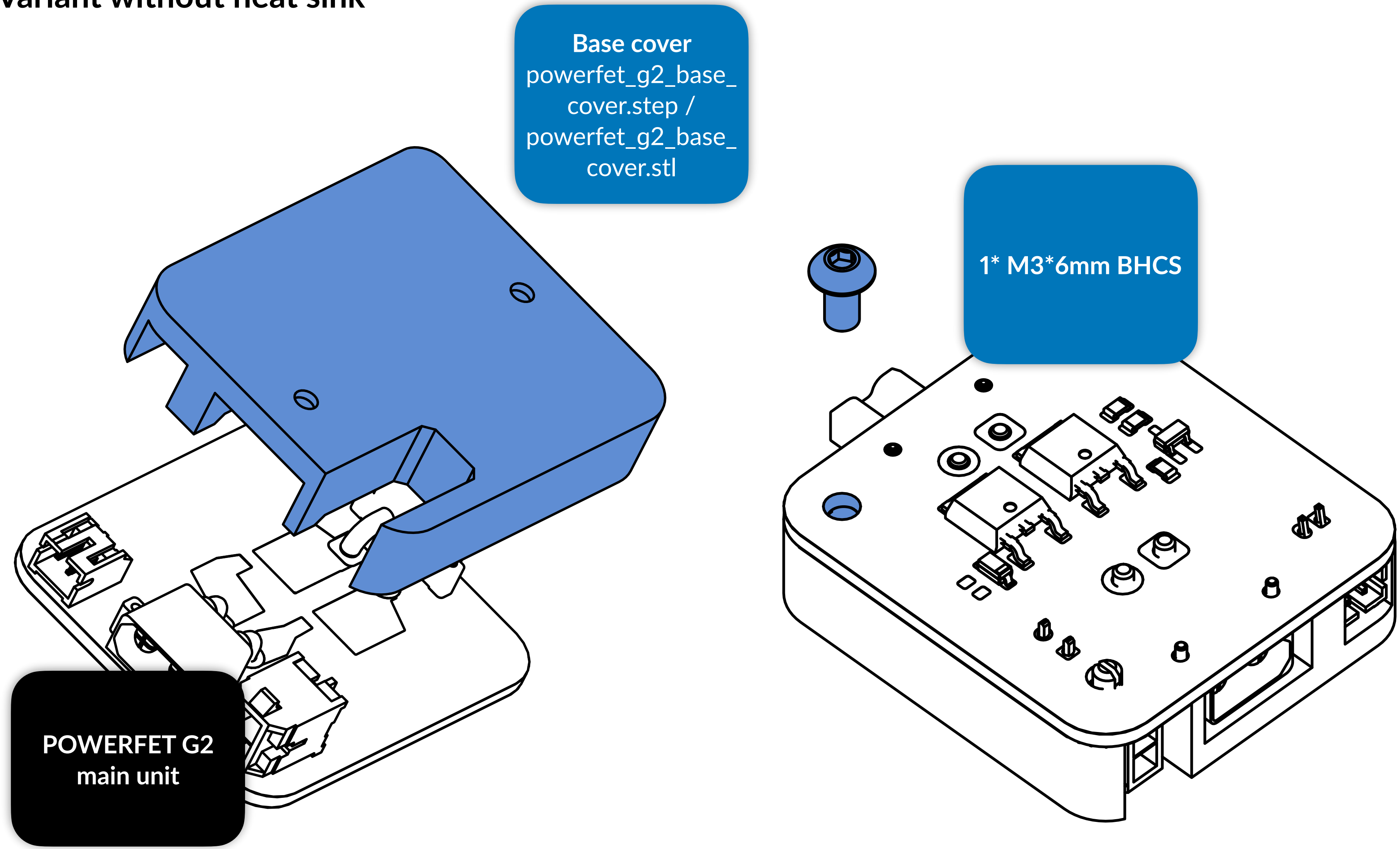
Example of solder bump on copper



Cover installation

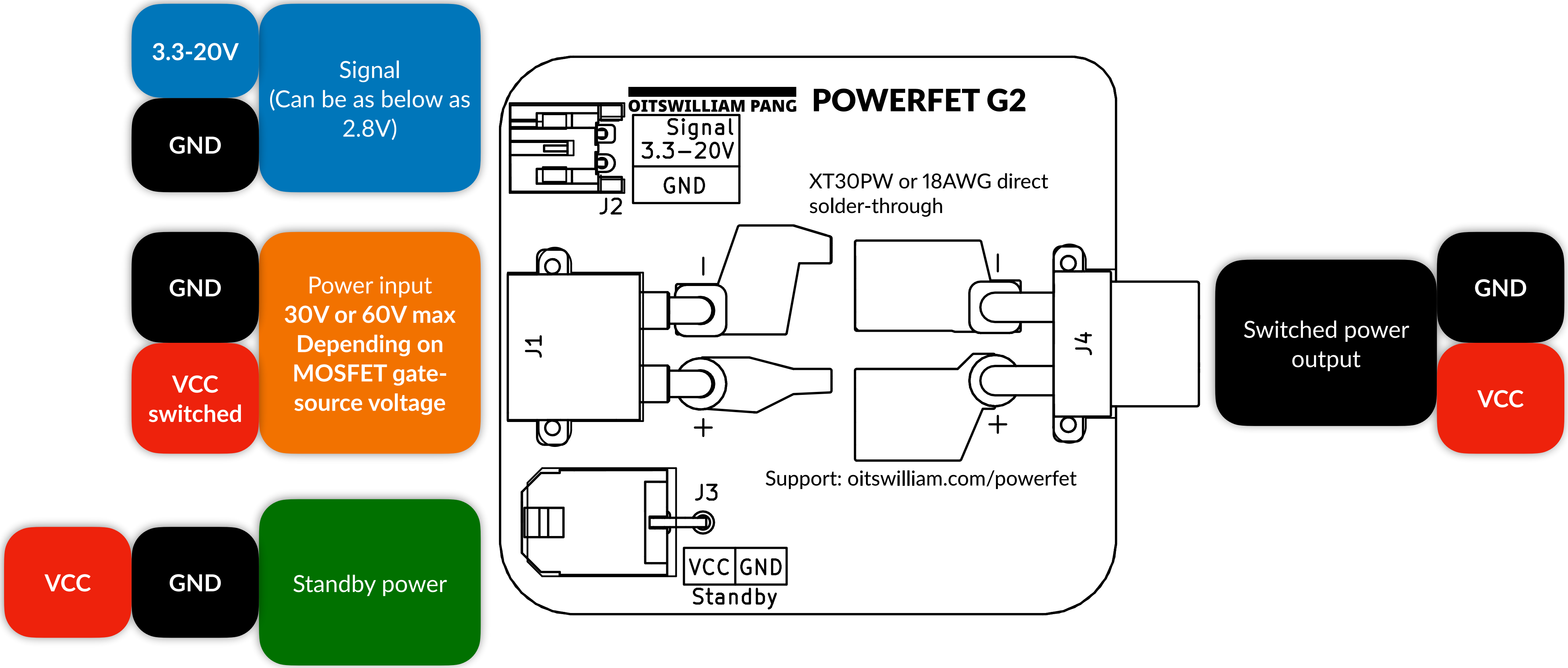
- The base covers for POWERFET G2 have holes that attach with DIN clips from Voron Design and secured via M2*10mm non-countersunk self-tapping screws.

Variant without heat sink



Cable connection

Pinouts - bottom view



Detailed explanation

- **Power input**
  - An XT30 male connector that connects from the DC power supply.
  - Maximum voltage varies by variant, normally either 30V, 60V or custom by user, depending on MOSFET gate-source voltage.
  - 15A typical, up to 70A of burst current (30V and with solder bump on copper).
  - Ground and power input VCC.
- **Switched power output**
  - An XT30 female receptacle that connects to the target device.
  - Switched by the dedicated signal port.
  - Switched ground and switched power output.
- **Standby power connector**
  - A Micro-Fit 3.0 2*1P connector that connects to standby components such as buck converters for host computers.
  - Always on, not being switched by any signal ports.
  - Power input VCC and ground.
- **Signal port**
  - A JST-PH 2.0 connector that detects DC output signal.
  - Detects voltage typically from 3.3 to 20V. It can detect as low as 2.8V. Using 5V as an example.
  - Signal input (min 2.8V, range 3.3 to 20V, example 5V) and ground.

Troubleshooting

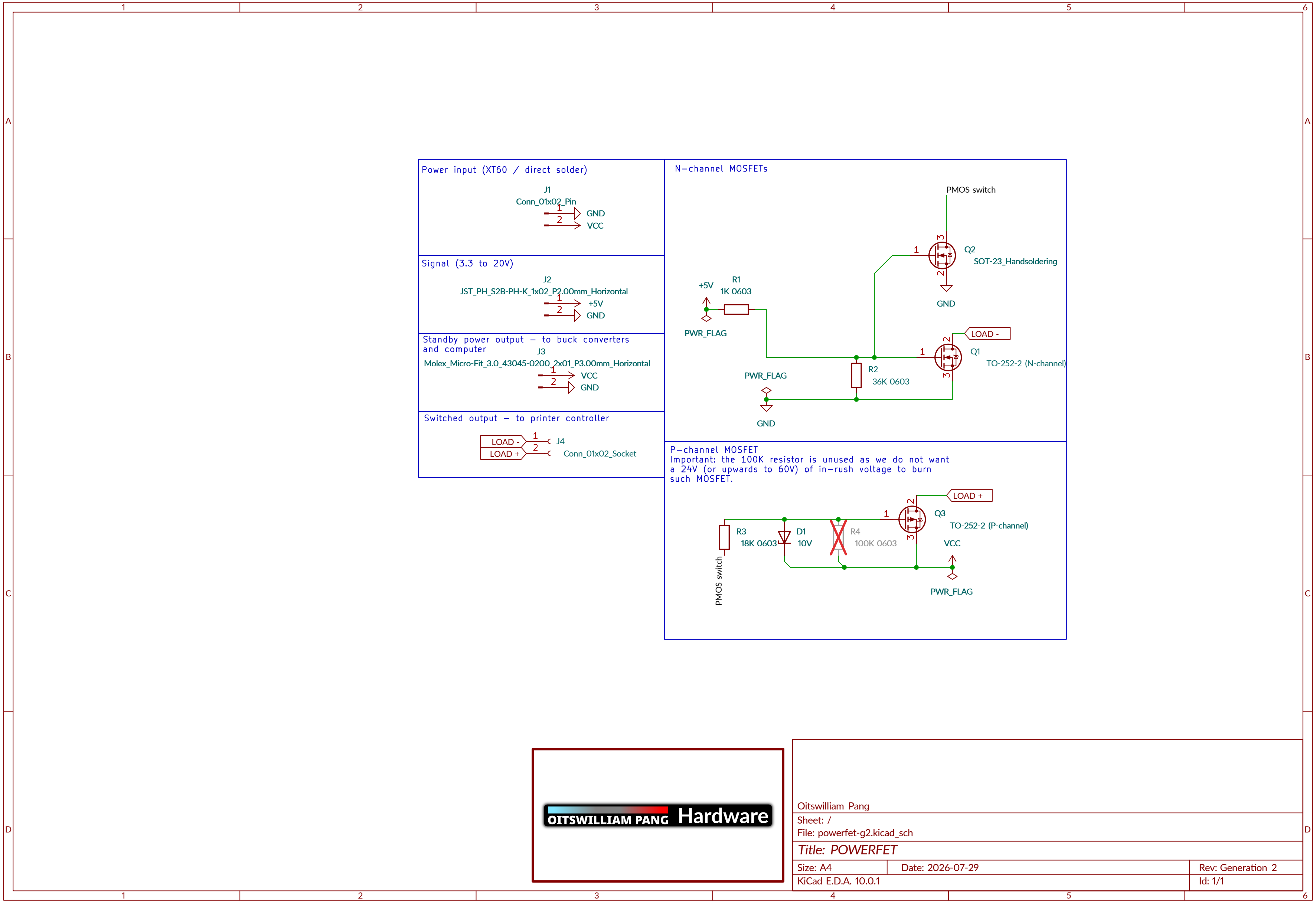
- The switch is always on! What should I do?
  - **End-user culprit:** the connection between the power input and the switched output is reversed. The XT30 male connector in POWERFET is to the power input, whereas XT30 female receptacle in it is to the switched output. Voltages under 20V will not cause damage.
  - Manufacturing culprit: there could a chance where non-guaranteed stores use inferior N-channel MOSFETs that are normally closed, or depletion-type for the MOSFET industry.
- R4 was in, should I occupy this?
  - In short answer, no, but in long answer, this was here for inserting a 100K-ohm resistor for voltage regulation, but since the Zener diode regulated the voltage, such is no longer required.

Assembly complete

And we are done! If you are satisfied with such product consider tagging @officialbunny350 on Facebook, @oitswilliam on Instagram or @realBunny350 on X (formerly known as Twitter) and share such product with us on Oitswilliam Pang Discord Server.

If you have enquires, problems or issues related to building or using such product, please report it on GitHub.com/Bunny350/OIWP-POWERFET and we will get on it.

Schematics



Power input (XT60 / direct solder)
J1
Conn_01x02_Pin
GND
VCC
Signal (3.3 to 20V)
J2
JST_PH_S2B-PH-K_1x02_P2.00mm_Horizontal
+5V
GND
Standby power output – to buck converters and computer
J3
Molex_Micro-Fit_3.0_43045-0200_2x01_P3.00mm_Horizontal
VCC
GND
Switched output – to printer controller
LOAD -
LOAD +
J4
Conn_01x02_Socket
N–channel MOSFETs
PMOS switch
Q2
SOT-23_Handsoldering
GND
+5V
R1
1K 0603
PWR_FLAG
LOAD -
Q1
TO-252-2 (N-channel)
R2
36K 0603
PWR_FLAG
GND
P–channel MOSFET
Important: the 100K resistor is unused as we do not want a 24V (or upwards to 60V) of in–rush voltage to burn such MOSFET.
LOAD +
Q3
TO-252-2 (P-channel)
R3
18K 0603
D1
10V
R4
100K 0603
VCC
PWR_FLAG
PMOS switch
OITSWILLIAM PANG Hardware
Oitswilliam Pang
Sheet: /
File: powerfet-g2.kicad_sch
Title: POWERFET
Size: A4
Date: 2026-07-29
Rev: Generation 2
KiCad E.D.A. 10.0.1
Id: 1/1

Change log

- 2026-08-20 - initial release

Manual designed by Oitswilliam Pang.
One-person operated design.
www.oitswilliam.com